PYTHON PROGRAMMING



21 Days Challenge

Day-18

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PYTHON MODULES, PACKAGES & LIBRARIES – WITH PIP



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What is a Modules

A Module is a single “.py” file that contains functions, variables, and classes.

Why use Modules?:

- Reuse code in multiple programs.
- Organize code into smaller, manageable files.

Create a Module

Save this code in a file named mymodule.py

```
def greeting(name):
  print("Hello, " + name)
```

Use a Module

Import the module named mymodule, and call the greeting function:

```
import mymodule


mymodule.greeting("Jonathan")
```

Python Built-in modules (math)

Mathematical functions and constants

```
import math


print("Square root:", math.sqrt(16))
print("Power:", math.pow(2, 3))
print("Pi value:", math.pi)
print("Factorial:", math.factorial(5))
```

Python Built-in modules (random)

Generating random numbers.

```
import random

print("Random number:", random.randint(1, 10))
print("Random float:", random.random())
print("Random choice:", random.choice(["apple", "banana", "cherry"]))
```

Python Built-in modules (datetime)

Working with dates and times.

```
import datetime


now = datetime.datetime.now()
print("Now:", now)
```

Python Built-in modules (os)

Interacting with the Operating System.

```
import os

print("Current directory:", os.getcwd())
print("Files:", os.listdir())

os.mkdir("test_folder")
os.rmdir("test_folder")
```

What is a Packages

A directory containing modules and a special __init__.py file.

Why use Packages ?:

- Structure large projects.
- Group related modules in one namespace.

How to Create and Access Packages

- Create a Directory
- Add Modules
- Include __init__.py
- Add Sub packages (Optional)
- Import Modules

```
Student/ (Package)
  | __init__.py
  | details.py
  | marks.py
  | collegeDetails.py
```

```
from Student import details
```

Python Built-in Packages

Internal packages → Always available, just import.

- sys → System-related functions
- json → Work with JSON
- re → Regular expressions
- http → Basic web requests
- xml → XML data

Python Built-in Packages

External packages → run “!pip install package_name" before using.

- NumPy → numerical computing
- Pandas → data analysis
- Matplotlib / Seaborn → data visualization
- OpenCV (cv2) → computer visioN

```
!pip install pandas
import pandas as pd
```

What is a Library

A Library is a collection of packages that provide ready-made tools and functionalities.

Why use Library :

- Save time (no need to write code from scratch).
- Provide powerful tools for tasks like data analysis, AI, and visualization.

Popular Libraries

- NumPy → numerical computing
- Pandas → data analysis
- Matplotlib / Seaborn → data visualization
- PyTorch, TensorFlow → machine learning
- Pygame → game development

QUESTION SESSION

THANK YOU

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